

**Advanced (Habanero)**

- Q1.** What is the discriminant in the quadratic equation  $x^2 + 5x + 6 = 0$ ?  
 (A) 1  
 (B) 13  
 (C) 25  
 (D) 13  
 (E) 29
- Q2.** Solve the quadratic equation  $x^2 + 2x - 6 = 0$  using the quadratic formula. What is  $x$ ?  
 (A)  $-1 \pm \sqrt{7}$   
 (B)  $-3 \pm \sqrt{2}$   
 (C)  $1 \pm \sqrt{3}$   
 (D)  $-1 \pm \sqrt{5}$   
 (E)  $-2 \pm \sqrt{3}$
- Q3.** For the quadratic equation  $ax^2 + bx + c = 0$ , what does the discriminant  $b^2 - 4ac$  determine?  
 (A) The number of solutions  
 (B) The type of solutions  
 (C) Both A and B  
 (D) The value of  $x$   
 (E) The value of  $y$
- Q4.** The quadratic equation  $x^2 - 4x - 5 = 0$  has how many solutions?  
 (A) 1  
 (B) 2  
 (C) 0  
 (D) -2  
 (E) -1
- Q5.** What is the discriminant of the quadratic equation  $x^2 + 10x + 24 = 0$ ?  
 (A) 16  
 (B) 4  
 (C) 16  
 (D) 64  
 (E) 24
- Q6.** The solutions to the equation  $2x^2 - 5x - 3 = 0$  are  $x = \frac{5 \pm \sqrt{61}}{4}$  and  $x = \frac{5 \pm \sqrt{61}}{2}$ . What is  $\boxed{61}$ ?  
 (A) The discriminant  
 (B) Twice the discriminant  
 (C) The value of  $x$   
 (D) The constant term  
 (E) None of the above
- Q7.** What is the nature of the solutions of  $3x^2 + 7x + 4 = 0$ ?  
 (A) Two distinct real solutions  
 (B) No real solutions  
 (C) Two equal real solutions  
 (D) No solutions  
 (E) Imaginary solutions
- Q8.** If  $b^2 - 4ac = 0$  for the quadratic equation  $ax^2 + bx + c = 0$ , then the equation has which type of solutions?  
 (A) Two distinct real solutions  
 (B) Two complex solutions  
 (C) One repeated real solution  
 (D) No real solutions  
 (E) No solutions

**Open Ended Questions (FRQ)**

- Q9.** Solve the quadratic equation  $x^2 - 7x + 10 = 0$  using the quadratic formula. Show your work and simplify your answer.
- Q10.** Determine the nature of the solutions of  $2x^2 + 3x + 1 = 0$ . Use the discriminant to justify your answer.